

L3: When Does $G(n,p)$ have a Single Connected Component?

Here is one more interesting question about the size of the LCC: **how large should p (or $\bar{k}$) be so that the LCC covers all network nodes?**

Note that the previous derivation did not answer this question – it simply told us that there is a phase transition when $\bar{k} = 1$.

Suppose again that S is the probability that a node belongs in the LCC.

Then, the probability that a node does NOT connect to ANY node in the LCC :

$$(1-p)^{Sn} \approx (1-p)^n \text{ if } S \approx 1.$$

The expected number of nodes not connecting to LCC:

$$\bar{k}_o = n \cdot (1-p)^n = n \left(1 - \frac{np}{n}\right)^n \approx n \cdot e^{-np}$$

Recall that $\left(1 - \frac{x}{n}\right)^n \approx e^{-x}$ when $x \ll n$. So we assume at this point of the derivation that the network is sparse ($p \ll 1$).

If we set $\bar{k}_o$ to less than one node, we get that:

$$n \cdot e^{-np} \leq 1$$

$$-np \leq \ln\left(\frac{1}{n}\right) = -\ln n$$

$$p \geq \frac{\ln n}{n}$$

$$\bar{k} = np \geq \ln n$$

which means that when **the average degree is higher than the natural logarithm of the network size ($\bar{k} > \ln n$) we expect to have a single connected component.**